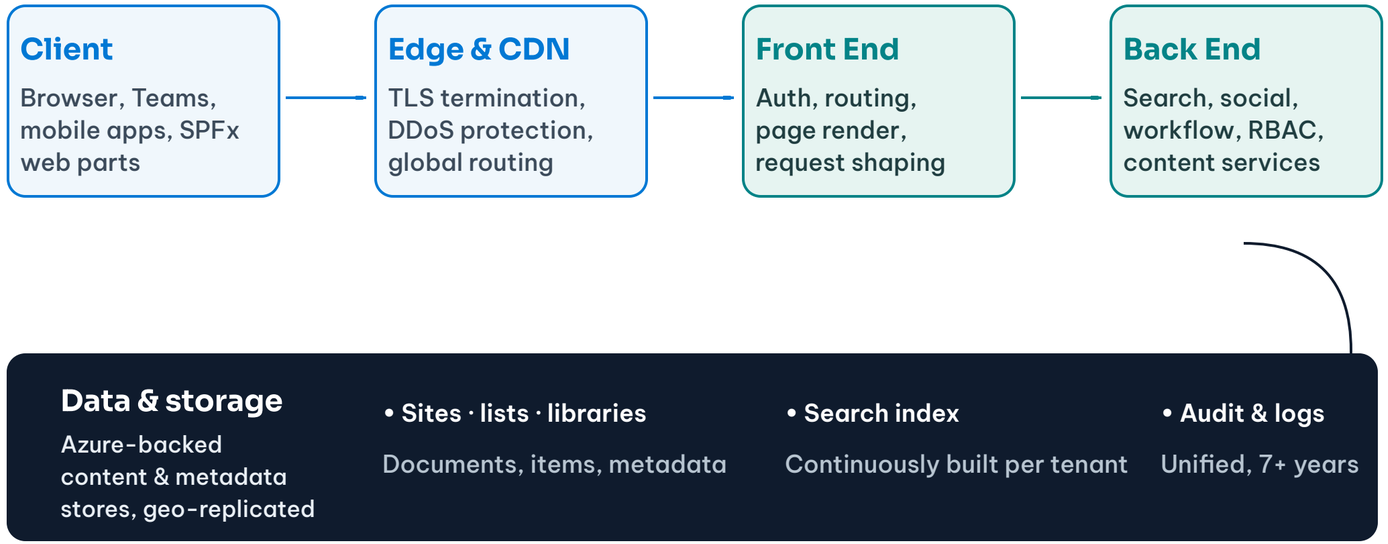
Service architecture overview

How a user request travels through SharePoint Online, from edge to back-end data plane.

Request journey



Key facts

Cloud PaaS, not servers
Microsoft manages patching, scaling and uptime.

Multi-tenant by default
Logical isolation inside a single Microsoft 365 tenant.

Auth via Microsoft Entra ID
Every call carries a bearer token issued by Entra.

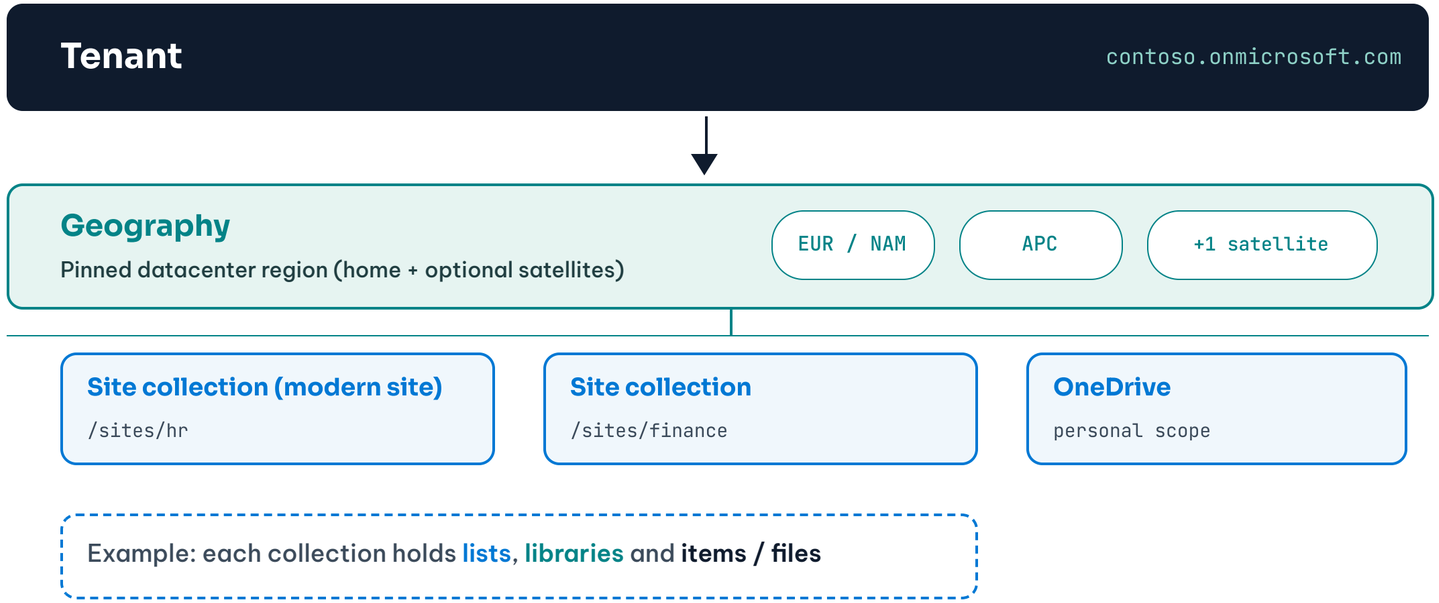
99.9%+ monthly SLA
Backed by Service Health & Microsoft 365 status feed.

```
// Diagram shows the request path, not every internal hop.
```

Tenant, geo & content hierarchy

How sites, lists and items live inside one tenant and are pinned to a geography for data residency.

Hierarchy inside a tenant



What to remember

- A **tenant** is one isolated Microsoft 365 workspace.
- A tenant has **one home geo** and optional **satellite** geos (Multi-Geo is a paid add-on).
- In modern SharePoint, **every site is its own site collection**: independent permissions, recycle bin, storage quota, navigation.
- **Default storage** per tenant is pooled; OneDrive and SharePoint share the budget.

// Geography rules data residency, not auth region.

SharePoint Online vs SharePoint Server

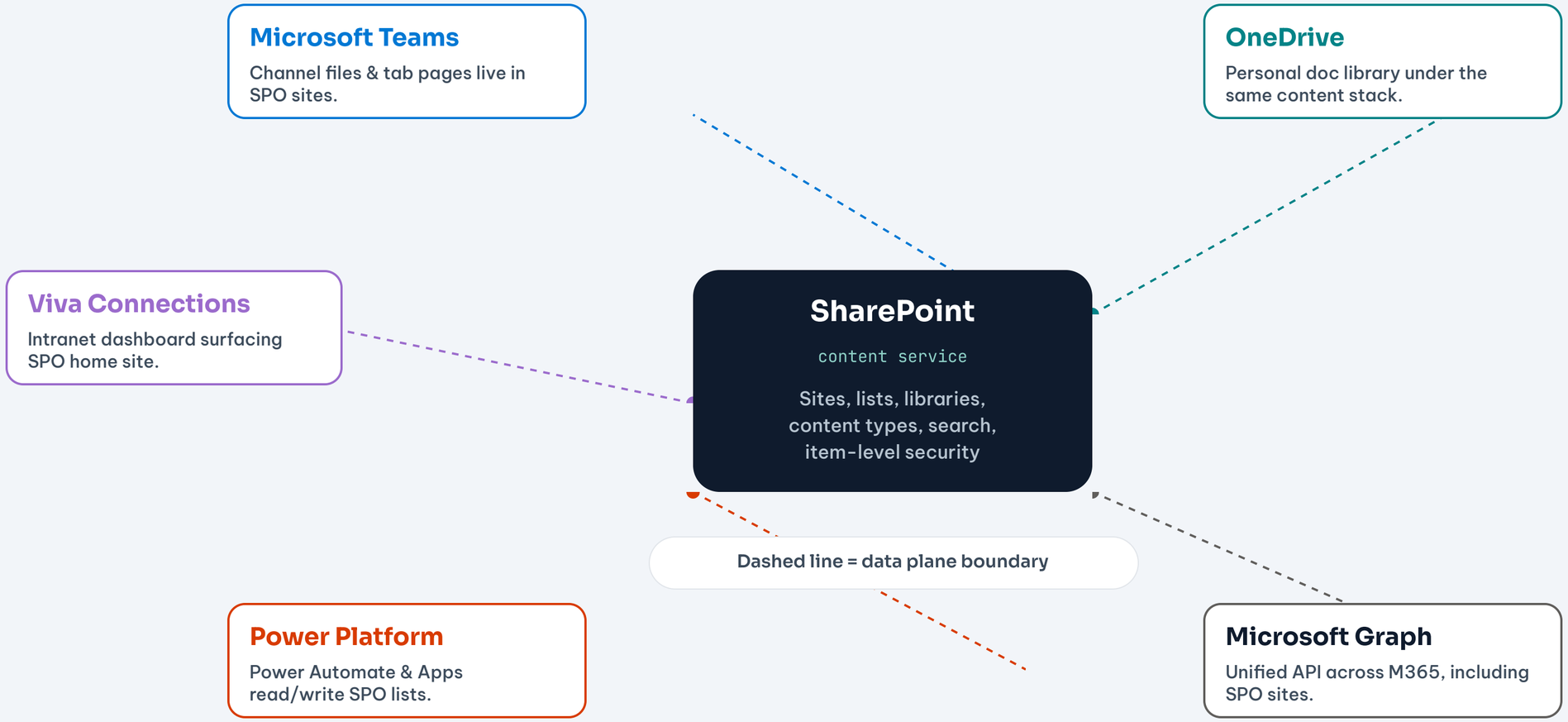
What changes when SharePoint moves from your datacenter into Microsoft 365.

Capability	SharePoint Online	SharePoint Server (Subscription)
Deployment Where it runs	Microsoft-operated cloud, per-tenant.	Self-hosted on customer VMs (Azure or on-prem).
Updates & cadence Who patches it	Microsoft pushes rolling changes, no admin action.	Cumulative updates you schedule; admin owns patching.
APIs & extensibility How you build on it	SPFx, REST, Microsoft Graph, Power Platform.	Same core APIs; classic farm solutions and WSPs also supported.
Hybrid story Bridges to other worlds	Connected to Teams, OneDrive, Viva, Power Platform by default.	Hybrid with SPO via Cloud Hybrid Search & SharePoint Hybrid.
When to pick it Practical trade-off	Default choice for new Microsoft 365 tenants.	Regulated industries with strict data residency needs.

// This deck focuses on SharePoint Online. Server-specific WSPs, Farm solutions and BCS are out of scope.

Where SharePoint fits in the Microsoft 365 ecosystem

SharePoint is the content service underneath Teams files, OneDrive, Viva, Power Platform and Graph.



Sites & site collections

In modern SharePoint the two collapse together: every modern site is a site collection.

Modern site = site collection

Team site

M365 group

`/sites/hr · Collaboration`

- Paired with a Microsoft 365 group
- Provisions a Teams team automatically
- Owners, members, visitors mapped to roles

Communication site

no group

`/sites/intranet · Broadcast`

- Read-heavy news, dashboards, home page
- Permissions only — no Teams team
- Often used as a hub or Viva landing

`// each box above is one full site collection — independent permissions, storage, recycle bin`

Inside one site collection you get

Lists

Structured rows with typed columns & forms. Issues, contacts, trackers, custom business data.

Libraries

Files + per-file metadata & versioning & co-authoring. Documents, images, videos, record libraries.

Pages & web parts

Modern pages host web parts such as News, Files, List, Highlighted Content, and your SPFx solutions.

Old vs modern model

Classic (legacy)

Site collection holds child sites, often deeply nested, with shared permission inheritance.

Modern (recommended)

Flat hierarchy. Each site is its own collection, joined by hubs, search and graph traversal.

Rule of thumb

Before creating a new site, ask: is the audience distinct, governance separate, or scope isolated? Otherwise, reuse a hub.

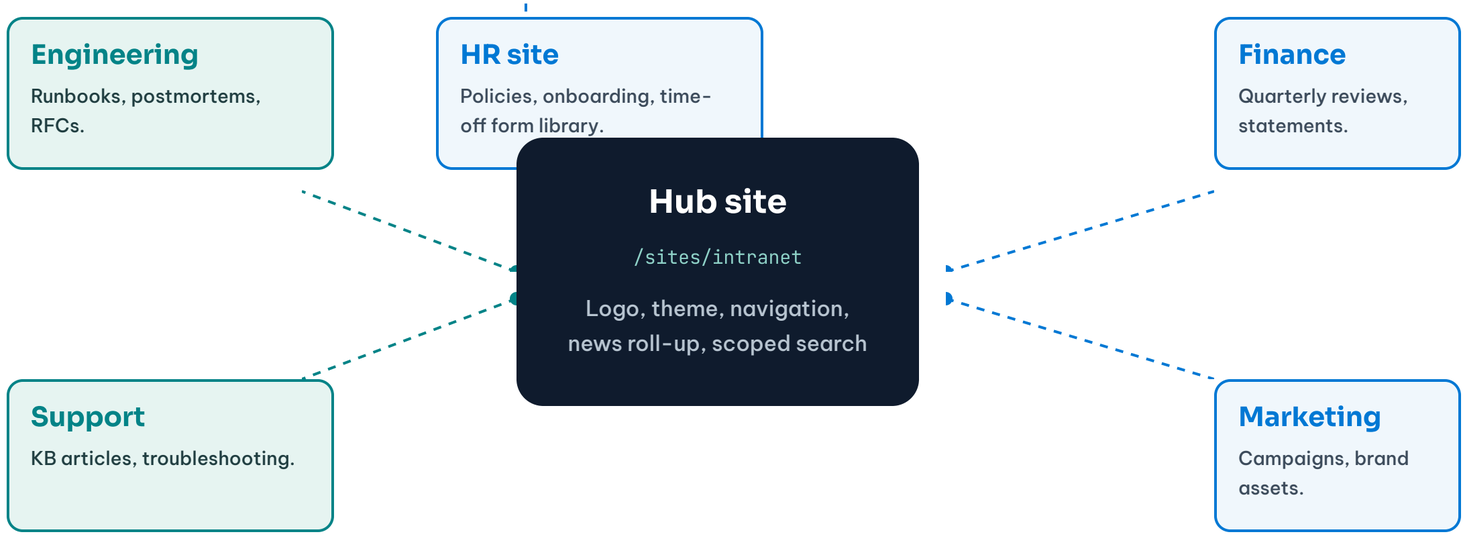
`URLs: /sites/<name>, /teams/<name>`

`Avoid: deep /sites/a/b/c paths.`

Hub sites & associations

A hub ties independent sites together by navigation, theme, search and shared permissions.

Anatomy of a hub




```
// one site can be associated with exactly one hub; hubs can be associated to one parent hub (multilevel)
```

What the hub does

Shared navigation
Top navigation and logo propagate.

Scoped search & news
Results filtered to the hub.

Hub-scoped permissions
Optional sync of membership to associated sites.

Common look & feel
Theme, header, footer applied to spokes.

```
Register-SPOHubSite
Add-SPOHubSiteAssociation -Site ...
```

Lists & libraries

Lists hold structured rows; libraries wrap files with metadata, versioning, and co-authoring.

Lists

Typed rows with columns, views, forms, alerts.

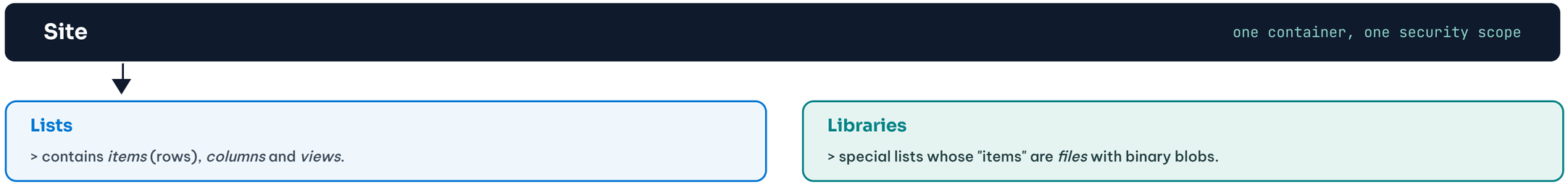
Title	Status	Owner	Due	Priority
Q3 onboarding plan	In progress	A. Garcia	11/14	High
Vendor review — ADP	Blocked	K. Patel	11/21	Med
Open enrollment FAQs	Drafting	R. Tanaka	12/01	Low
Annual review template	Done	M. Lopez	10/30	Med

Libraries

Files + per-file metadata + versioning + checkout.

Name	Modified	Modified By	Version
Handbook-2026.docx	10/24	a.garcia@	v12.4
Org-chart.png	10/19	r.tanaka@	v3.0
Roadmap.pptx	10/22	m.lopez@	v7.1
Vendor MSA.pdf	10/17	legal-team@	v2.0

Where they live in the data model



Columns & content types

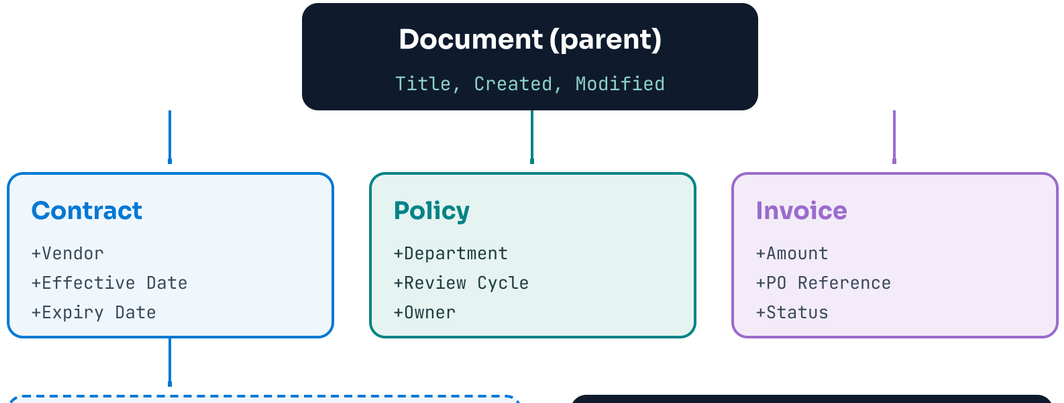
Columns describe fields, content types package them into reusable "shapes" that lists and libraries inherit from.

Common column types

Single line of text	`string, indexed for search`
Choice	`dropdown, multi-choice allowed`
Number / Currency	`typed numeric with culture`
Date / Time	`date-only or with time`
Person / Group	`Entra lookup, single or multi`
Lookup	`cross-list reference, optional expand`
Managed metadata	`Term Store backed, hierarchical`

...also: Yes/No, Hyperlink, Image, Location, Geolocation, calculated

Content type inheritance

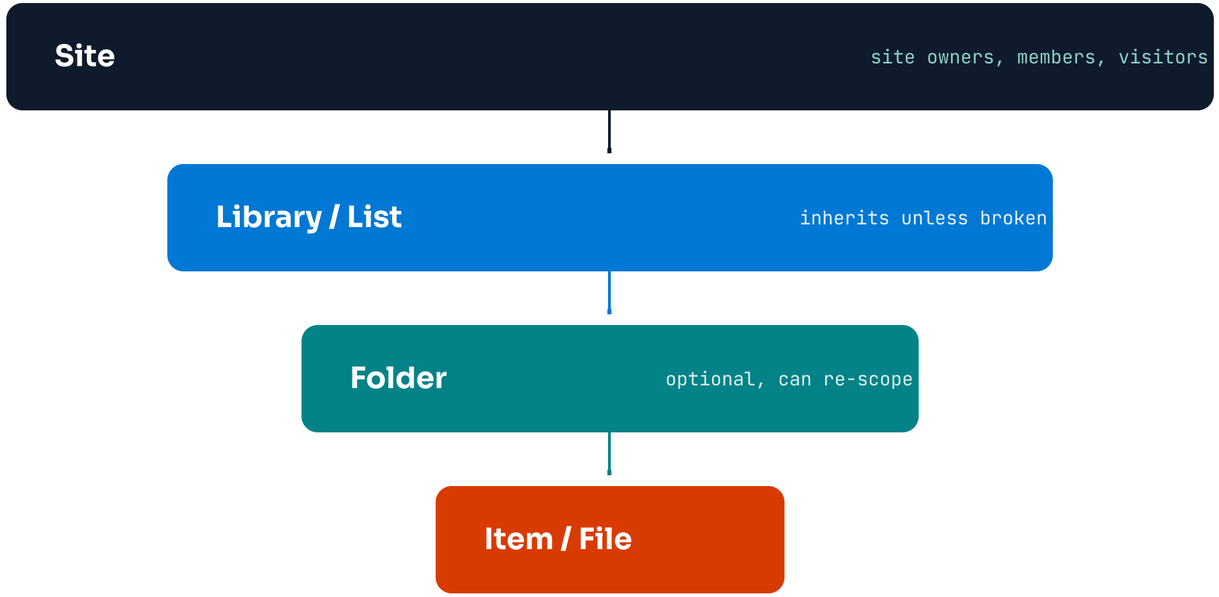



```
// A library can host many content types.
// New file uploads pick one — drives required columns.
Add-PnPContentType -Name "Contract"
  -Parent "Document"
```

Permission levels & inheritance

Permissions cascade top-down by default. Each level can override its parent only when inheritance is broken.

Inheritance cascade



// Each box is a securable scope; permissions can be set or broken at any of these.

Default permission levels

Full Control
All rights, including change permissions.

Design
Add/change lists, lists and pages, but not permissions.

Contribute
Add, edit, delete items — the everyday author level.

Read
View pages, items — default for visitors.

View Only
Cannot see what they cannot open; used for audit-only audiences.

Levels are bundles of ~30 granular permissions such as Read, Write, Delete, ManageLists.

Breaking inheritance & groups

When default inheritance is the wrong answer: scope re-grants, then choose Entra groups or SharePoint groups.

Stop the cascade

1. Default — inherits

Library «HR Confidential» sits under site «HR». Everyone who can edit HR can also re-edit the library — including the sensitive offer letters list. Same ACL all the way down.

→ break

2. Stop, re-grant

Only HR-Admins group has Contribute; HR-Readers gets Read; visitors get View Only. Site-wide access no longer leaks. Unique ACL explicit & auditable.

Entra vs SharePoint groups

Microsoft Entra security groups

Tenant-wide, governed in Entra ID, work everywhere in M365 (Teams, Outlook, Defender).

SharePoint groups (legacy)

Site-scoped, owned by that collection. Owner-led, harder to audit at scale.

Best-practice checklist

- Prefer Entra groups over SharePoint groups — one membership source.
- Avoid unique permissions per item — they explode ACL size.
- Restrict «Full Control» to a named admins group, not «Site Owners» blindly.
- Set site access requests off by default at scale.

```
Set-PnPListPermission -List "Offer Letters" -Identity   "c:0-.f|rolemanager|contoso\HR-Admins" -AddRole   Contribute
Set-PnPGroup -Identity "HR-Readers" -SetRoleAs Read
```

SharePoint REST API fundamentals

A shape-shifting endpoint that mirrors the CSOM object model, anchored at `_api/`.

Endpoint anatomy

```
https://contoso.sharepoint.com/sites/hr/_api/web/lists
```

scheme	host + path	marker	resource
HTTPS, always.	Site host + /sites/<name>	/_api selects the REST surface	web, lists, sites, users ...

OData query options

```
$select=Title,Status   $filter=Status eq 'Active'   $orderby=Modified desc
$top=50   $skip=100   $expand=Author/Title   $count=true
```

verbs: GET (read), POST (create), MERGE/PATCH (partial update), PUT (replace), DELETE

POST/PATCH must send a `X-RequestDigest` from `contextinfo`, plus JSON body.

Tip: keep server-driven paging. REST endpoints cap rows-per-call (~5000 items, ~100 per page by default) — always watch `__next`.

Auth, headers, content type

```
Authorization: Bearer {token-from-Entra}
```

```
Accept: application/json;odata=verbose
```

```
X-RequestDigest: {form-digest-value}
```

Modern SDKs (PnPjs, PnP PowerShell) hide these for you. Good to know the headers before automating.

REST request & response examples

Three shapes a developer writes every day: get-list, get-items with $expand, create-item with POST.

GET — read list metadata

```
GET https://contoso.sharepoint.com/sites/hr/_api/web/lists?
$select=Title,ItemCount,BaseTemplate
Accept: application/json;odata=verbose
```

POST — create an item

```
POST /sites/hr/_api/web/lists/getbytitle('Projects')/items
Content-Type: application/json;odata=verbose
X-RequestDigest: {digest}
{ "Title": "Q3 Hiring Plan", "Status": "Drafting" }
```

PATCH partial update

```
POST .../items(7)/validateUpdateListItem
If-Match: *
{ "formValues":[{"FieldName":"Status","FieldValue":"Done"}] }
```

Sample JSON response

```
{
  "d": {
    "results": [
      {
        "Title": "Q3 Hiring Plan",
        "Id": 7,
        "GUID": "c5e9...",
        "Status": "Drafting",
        "AuthorId": 12,
        "EditorId": 8,
        "Modified": "2026-10-23T...",
        "__metadata": {
          "uri":".../items(7)",
          "type":"SP.Data.ProjectsItem"
      }
    }
    ...
    ],
    "__next": "...?$skiptoken=PAGE2"
  }
}
```

Wrap ‘verbose’ mode or use `SP.SOM.Lists`+odata=nometadata in apps where payload size matters.

Introduction to Microsoft Graph

One endpoint, one auth model, one SDK that reaches SharePoint, OneDrive, Outlook, Teams, Identity and more.

Endpoint anatomy

```
GET https://graph.microsoft.com/v1.0/sites/{site-id}/lists

// same shape, plus me, users, groups, drives, teams, chats, identity — resource graph.
```

Why developers reach for Graph

Single token across M365

One bearer token reads data in SharePoint, Outlook mail, Teams chat, Entra ID.

Cross-resource queries

$expand from a user to their manager, joined groups, drive items.

First-class SDKs and tools

SDKs for .NET, JS, Python, Go, Java, plus Graph Explorer in the browser.

Webhooks & change notifications

Subscribe to item / list / drive changes, get delta queries for sync.

Calls you will write

`GET /v1.0/sites/contoso.sharepoint.com:/sites/hr`
resolves a site by host + path.

`GET /v1.0/sites/{id}/lists?$filter=displayName eq 'Projects'`
targets a specific list with OData filter.

`POST /v1.0/sites/{id}/lists/{id}/items`
creates an item with a JSON body.

`GET /v1.0/sites/{id}/lists/{id}/items/delta`
sync incremental changes since a token.

`POST /v1.0/subscriptions { resource, changeType, notificationUrl }`
registers a webhook to receive notifications.

Delta query, throttling, scopes

Sync efficiently, stay under limits, ask for the smallest scope your scenario actually needs.

Delta query flow




```
"@odata.deltaLink": "/sites/.../items/delta?$deltatoken=..."
"value": [{ "id":"..." ...}, "@removed": { "reason":"deleted" } ]
```

Throttling in plain numbers

4 000
requests / 20s
per tenant, all apps

HTTP 429
the signal you are pushing too hard

≤ 20
ops per JSON-batch request

Delegated permission scopes

Scope	What it allows
`Sites.Read.All`	Read sites in all site collections.
`Sites.ReadWrite.All`	Edit items across all sites.
`Files.ReadWrite`	Read & write user's own files.
`Group.ReadWrite.All`	Manage M365 groups & memberships.
`User.ReadBasic.All`	Read basic profile for all users.

Honour Retry-After

`429` responses carry a `Retry-After` header. Honour it before retrying.

Add jitter, batch with `$batch`, cache people lookups, never tight-loop on retries.

The developer tooling stack

Six tools every trainee should leave the room knowing by name and shape.

V Visual Studio Code

Editor & SPFx bundler

Free, cross-platform. Install SPFx, REST Client, Graph extensions & paste JWT tokens with ease.

S SharePoint Framework

Web parts, extensions, libs

```
npm i -g yo @microsoft/sharepoint
yo @microsoft/sharepoint
gulp serve --nobrowser
```

N Node.js LTS

Runtime baseline

SPFx 1.21+ supports Node 22 LTS. v24 and v26 in roadmap. Use nvm on Windows/macOS.

P PnP PowerShell

Provisioning & admin

```
Install-Module PnP.PowerShell
Connect-PnPOnline -Url ... -Interactive
```

G Graph Explorer

Browser playground

developer.microsoft.com/graph/graph-explorer
Sign in with your dev tenant to test live calls.

M M365 Developer Tenant

Free sandbox

developer.microsoft.com/microsoft-365
90-day renewable, with sample data packs.

Also worth knowing

- PnPjs — fluent JS wrapper around REST + Graph
- CLI for Microsoft 365 — cross-platform shell scripting
- Postman — HTTP client with Graph + REST collections

```
npm i @pnp/graph @pnp/sp @azure/msal-node
npm i -g @pnp/cli-microsoft365
# Postman: import microsoft-graph.postman
```

Environment setup flow

A repeatable checklist from a fresh laptop to a working SPFx + Graph loop.

1 Claim a dev tenant

Join the Microsoft 365 Developer Program.
Add a sample data pack so you don't start empty.

2 Install Node + VS Code

```
nvm install 22
nvm use 22
# install VS Code & Node 22 LTS
```

3 Install SPFx generator

```
npm i -g yo gulp-cli
npm i -g @microsoft/sharepoint
```

4 Install PnP PowerShell

```
Install-Module PnP.PowerShell
Connect-PnPOnline -Url ... -Interactive
```

5 Scaffold a web part

```
yo @microsoft/sharepoint
gulp trust-dev-cert
gulp serve
```

6 Verify with Graph Explorer

Sign in with dev tenant. Hit `/me`, then `/sites`.

Done when

- `node --version` prints v22
- `yo --version` returns a version number
- `Get-Module PnP.PowerShell -ListAvailable` works
- A dev-loaded web part shows in the Workbench

Common pitfalls

Node version mismatch can break gulp. Disable browser proxy. Trust the dev cert only locally. Use a real upn, not a guest account, while prototyping.

Lab Part 1 — site, lists, libraries

Provision the «HR Lite» workspace end-to-end with PnP PowerShell. Estimated time: 30-40 min.

1 Connect to your dev tenant

```
Connect-PnPOnline -Url "https://contoso.sharepoint.com" -Interactive
Get-PnPContext  ·  confirms the connection.
```

2 Create the team site

```
New-PnPSite -Type TeamSite -Title "HR Lite" -Alias "hrlite"
-Description "HR workspace for the dev tenant"
```

3 Add «Projects» list

```
Add-PnPList -Title "Projects" -Template GenericList
Add-PnPField -List "Projects" -DisplayName "Status" -Type Choice
```

4 Add «Policies» library

```
Add-PnPList -Title "Policies" -Template DocumentLibrary
Enable-PnPFeature -Identity DocId -Scope Site
```

5 Seed four items via REST

```
Use Invoke-PnPSPRest or Postman with items endpoint
Try $select, $filter so you see real payload behavior
```

6 Verify in browser

Open /sites/hrlite, expand *Site contents* — confirm Projects list and Policies library exist.

Expected results after step 6

• One team site under /sites/hrlite · • Projects list with Status column · • Policies doc library · • Four seeded items visible to read-only users

Lab Part 2 — columns, types, security

Build a content type, attach it, then scope one library for a smaller audience.

Build the «Contract» content type

a — Add a parent child of Document

```
Add-PnPContentType -Name "Contract" -Parent "Document" `
  -ContentTypeId "0x0101ABCD..."
```

b — Add fields Vendor, EffectiveDate, ExpiryDate

```
Add-PnPField -DisplayName "Vendor" -Type Text -AddToDefaultView
Add-PnPField -DisplayName "EffectiveDate" -Type DateTime
```

c — Attach «Contract» to Policies library

```
Add-PnPContentTypeToList -List "Policies" -ContentType "Contract"
```

d — Upload sample · check the columns

New documents now ask for Vendor, EffectiveDate, ExpiryDate — the typed contract lives.

Tighter scope on «Contracts»

In PART 2 we narrow access on this one library only.

e. Break inheritance

```
Set-PnPList -Identity "Contracts" `
  -BreakRoleInheritance -CopyRoleAssignments
```

f. Add an Entra group

Provision "HR-Readers" Entra group and add two test users.

g. Grant per role

```
Set-PnPListPermission ... -AddRole
  Contribute, Read, ViewOnly
```

h. Validate with a guest

Sign in as a non-member — Projects visible, Policies visible, Contracts visible only to HR-Admins. Audit log under Purview

Three industry scenarios

How Day 1's parts come together in real projects — retail, financial services, HR operations.

Retail · Hub migration

Contoso Retail, 4 200 staff

Migrate 19 region sites into a single hub with department spokes. H&M hierarchy simplifies per-store bulletins and compliance training material.

Stack: PnP PowerShell + Graph delta + SPFx web part for emergency banner.

Approach

1. map territory with Graph delta of /sites
2. promote one old site as hub via REST
3. associate spokes in waves over weekends
4. drill-down news roll-up runs on hub scope

Outcome: 19 found, 1 hub, 18 spokes, 0 redirect errors.

Bank · Document governance

NorthBank, regulated

Enforce retention, sensitivity labels and absolute uniqueness on contract PDFs across the legal library.

Stack: content types + sensitivity + retention labels + audit.

Approach

1. create a «Legal Contract» content type
2. apply «Highly Confidential» label
3. attach a 7-year retention policy
4. restrict Edit to a single legal team

Outcome: 100% of new contracts auto-labelled.

HR · Onboarding automation

Acme Corp, 1 800 hires / yr

When HR approves a new hire in Workday, a flow provisions a personal OneDrive, a Team site, an inbox, and bookmarks to policy docs in the HR hub.

Stack: Power Automate + Graph + Terraform-style site templates via PnP.

Approach

1. Workday webhooks → Power Automate
2. POST /users + /sites on success
3. copy PnP template to new site
4. broadcast welcome email via Graph

Outcome: 14-minute onboarding, no manual tickets.

Day 1 recap & Day 2 preview

Five things you should remember — and four topics that arrive next.

Five takeaways

1. **SPO is a tenant service, not servers**
   Microsoft operates the farm. Your job is governance and building on top.
2. **Modern site = site collection**
   Flat hierarchy, joined via hubs — not deep /sites/a/b/c nesting.
3. **Prefer Entra groups over SP groups**
   One membership; works across M365; auditable in Entra.
4. **Use REST for SPO nuance, Graph for cross-M365**
   Two API surfaces, one identity model. Pick the closer fit. Honour 429.
5. **Provision with PnP, extend with SPFx, automate with Power Platform**
   Three layers, one story. Day 2 climbs the third.

Day 2 in four blocks

Tomorrow we move from data to behaviour.

Block 1 · Deep SPFx dive
Web parts, application customizers, library extensions, deployment via tenant app catalog.

Block 2 · Power Platform & SPO
Power Automate flows, custom connectors, Power Apps forms bound to lists.

Block 3 · Search & content intelligence
SharePoint search schema, query variables, vertical results, Graph connectors.

Block 4 · ALM & ALM-ready
Source control, dev/test/prod promotion, versioned SPFx packages, governance checklist.